

L9a: Introduction to Derivatives and European Option Pricing

CHEME 5660 Cornell University

Copyright Jeffrey Varner 2026. All Rights Reserved

Disclaimer and Risks

Training only. This content is for training and informational purposes only. The teaching team does not make, give, or endorse any offer or solicitation to buy or sell securities or derivative products, or provide investment or trading advice or strategy.

Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

We introduce derivatives and begin our discussion of derivatives pricing. What is a payment that may never arrive worth today?

Today's concepts

- Define and classify derivatives: futures, forwards, swaps, and options, with their underlying assets and market structures.
- Analyze option payoffs and profits: payoff and profit for long and short calls and puts, and how the strike and premium affect profitability at expiration.
- Price European options contracts: the premium makes the expected net present value zero under the risk-neutral pricing rule. We compute it by Monte Carlo simulation and with the Black–Scholes–Merton formulas, and relate calls and puts with put–call parity.

Lectures and Examples

Lecture:
CHEME-5660-L9a-Lecture-IntroductionToDerivatives-BSM-Fall-2026.ipynb

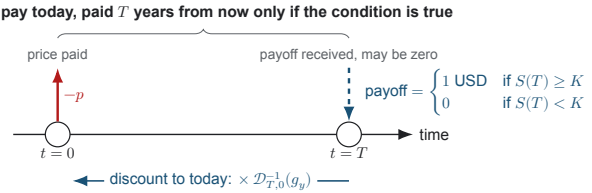
Example 1:
CHEME-5660-L9a-Example-SingleContractPayoffProfit-Fall-2026.ipynb

Example 2:
CHEME-5660-L9a-Example-BSM-Premium-Fall-2026.ipynb

The first example computes payoff and profit for single long and short call and put contracts. The second computes European call and put premiums by Monte Carlo simulation and with the Black–Scholes–Merton formulas, then looks at how the premium changes with strike price, in the model and in real S&P 500 index option quotes.

Concept Review: A Bet on the Future

Suppose we pay p today for a contract that pays 1 USD in T years if the share price $S(T)$ finishes at or above a target share price K, and nothing otherwise.



Discounting the payoff $V(S(T))$ at the benchmark growth rate g_y, with $\mathcal{D}_{T,0}^{-1}(g_y) = e^{-g_y T}$, gives a random NPV:

$$\text{NPV}_0 = -p + \mathcal{D}_{T,0}^{-1}(g_y)\, V(S(T)) = \begin{cases} -p + e^{-g_y T} & \text{if } S(T) \geq K \text{ (true)} \\ -p & \text{if } S(T) < K \text{ (false)} \end{cases}$$

Expected NPV of the Bet

Under our beliefs $\mathbb{P}$ about $S(T)$, the expected NPV is:

$$\mathbb{E}[\mathrm{NPV}_0] = (-p + e^{-g_y T}) \underbrace{\mathbb{P}(S(T) \geq K)}_{\text{true}} + (-p) \underbrace{\mathbb{P}(S(T) < K)}_{\text{false}}$$

weight by probability

$$= -p \underbrace{[\mathbb{P}(S(T) \geq K) + \mathbb{P}(S(T) < K)]}_{=1} + e^{-g_y T} \mathbb{P}(S(T) \geq K)$$

collect the p terms, probabilities sum to one

$$= -p + e^{-g_y T} \underbrace{\mathbb{P}(S(T) \geq K)}_{\text{true}}$$

The expected NPV is the discounted expected payoff minus the price paid today.

The Bet Is a Derivative Contract

The bet has a non-negative expected NPV when its price is at or below its discounted expected payoff:

$$p \leq e^{-g_y T}\, \mathbb{P}(S(T) \geq K)$$

The bet is a derivative contract: its price and payoff depend on the future price of another underlying asset (or event).

Each contract trades at a market price that every buyer pays, yet buyers with different beliefs $\mathbb{P}$ value the same contract differently.

How is the market price set?

Company Profile: Cboe Global Markets

Cboe operates markets for options, equities, futures, and foreign exchange. It opened on April 26, 1973, as the Chicago Board Options Exchange, the world's first exchange for listed options.

- Listed options: standardized strike prices, expiration dates, and contract sizes, so the same contract can trade many times a day. The Options Clearing Corporation (OCC) stands between the buyer and the seller, so if one side cannot pay, the other is still paid.
- Index options and volatility: SPX options are European-style and cash-settled. The VIX reads the market's expected 30-day volatility of the S&P 500 from their premiums.

European-style index options such as SPX are the contracts we price later in this lecture.

Company Profile: Cboe Event Contracts

In June 2026, Cboe began listing event contracts, binary options on the Mini-S&P 500 index:

- Payoff: a yes contract pays 100 USD if the index settles at or above a set level, and nothing otherwise.
- Clearing: OCC clears it like any other listed option.
- Connection to today: it is the bet from the Concept Review, paying 100 USD instead of 1 USD.

Cboe is the largest options exchange operator in the United States, and most contracts traded on it are calls and puts. Whatever the type of contract, each trades at a single premium that all buyers pay today. **Where does that number come from?**

What Is a Derivative?

A derivative is a contract between a buyer and a seller to transact an asset at a set price on a future date. It is an abstract asset whose cash flows depend on an underlying asset or event.

- Futures, forwards, and swaps: obligate both parties to transact. Futures trade on exchanges with standard terms. Forwards trade over the counter, where the two parties negotiate the terms, and if one party cannot pay, the other takes the loss. Swaps exchange one asset's cash flows for another's, also over the counter.
- Options: the buyer holds the right, but not the obligation, to buy (call) or sell (put) at a set price. Stock, index, and ETF options are exchange-traded.
- Event contracts: pay a fixed amount, usually one dollar, if an event occurs and nothing otherwise. Kalshi lists them.

We focus on options in this lecture.

How Big Is the US Options Market?

The U.S. options market is large, and it is still growing. Measured by notional value, the value of the underlying assets the contracts cover, more value trades each day in options than in stocks:

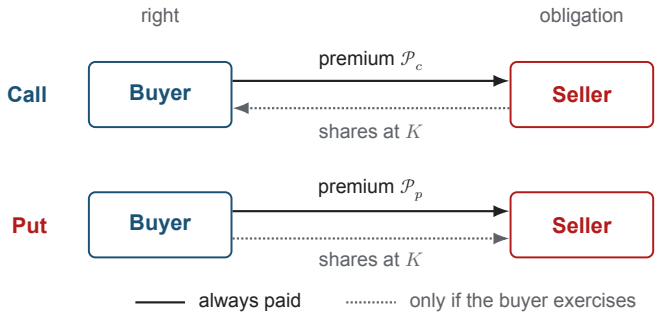
- Contracts traded: a record 15.2 billion in 2025, about 60.6 million per day. Through September 2026, the daily average rose to about 70.6 million contracts.
- Notional traded: about 4 trillion USD per day in 2025, more than three times the 1.1 trillion USD per day traded in U.S. equities.
- Notional value at one expiration: a record 6.6 trillion USD of options were set to expire on December 20, 2024, a quarterly expiration when stock options, index options, and index futures all expire.

Most of this volume is calls and puts, the contracts we price in this lecture.

Sources: OCC, Traders Magazine, Cboe, and MarketWatch; the lecture notebook's market table has the dates and links.

Call and Put Options Contracts

The two types of options contracts are calls and puts. In each, the buyer holds the right and the seller holds the obligation:



The buyer always pays the premium. The shares move at the strike price only if the buyer exercises.

The Business Case for Long and Short Calls

A call gives the owner the right, but not the obligation, to buy 100 shares (per contract) from the seller at the strike price K by the expiration date. What is the business case for each side?

- Buyer (long): can benefit from upside price movement without buying the shares. If the buyer exercises, they buy shares at K instead of the market price. The most the buyer can lose is the premium.
- Seller (short): receives the premium and must sell shares at K if the call is exercised. Calls are often sold by investors who already own the shares, 100 per contract (a covered call).

The buyer's risk is limited to the premium. The seller's risk has no upper limit, because the share price can rise without bound.

The Business Case for Long and Short Puts

A put gives the owner the right, but not the obligation, to sell 100 shares (per contract) to the seller at the strike price K by the expiration date. What is the business case for each side?

- Buyer (long): can benefit from downside price movement without selling the shares. If the buyer exercises, they sell shares at K instead of the market price. The most the buyer can lose is the premium.
- Seller (short): receives the premium and must buy shares at K if the put is exercised. Puts are often sold by investors with the cash to buy the shares, 100 per contract (a cash-secured put).

The buyer's risk is limited to the premium. The seller's risk is large but bounded: at a share price of zero, the seller loses K minus the premium's future value per share.

Call and Put Payoffs

The payoff per share of a single call or put contract at expiration (or when the contract is exercised) is:

$$\boxed{\begin{aligned} V_c(K, S(T)) &= \max(S(T) - K,\ 0) = (S(T) - K)^+ \\ V_p(K, S(T)) &= \max(K - S(T),\ 0) = (K - S(T))^+ \end{aligned}}$$

where K is the strike price and $S(T)$ is the share price at expiration (time $t = T$). The seller charges the buyer a premium per share, $\mathcal{P}_c(K, S(0))$ for a call and $\mathcal{P}_p(K, S(0))$ for a put.

The premium is paid at $t = 0$ and the payoff arrives at $t = T$, so we compare them at expiration using the premium's future value $\mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}$, where $\mathcal{D}_{T,0}(g_y) = e^{g_y T}$.

Profit at Expiration

The buyer's profit per share for a single contract at expiration is the payoff minus the premium's future value:

$$P_c^{\text{buyer}}(K, S(T)) = V_c(K, S(T)) - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c(K, S(0))$$
$$P_p^{\text{buyer}}(K, S(T)) = V_p(K, S(T)) - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p(K, S(0))$$

A position's orientation θ is $+1$ for the buyer (long) and -1 for the seller (short). Its profit per share is $\theta \cdot P^{\text{buyer}}$, so $P^{\text{seller}} = (-1) \cdot P^{\text{buyer}}$ and $P^{\text{buyer}} + P^{\text{seller}} = 0$ at every share price.

Options are a zero-sum game: one party's gain is the other's loss.

Breakeven

Setting the buyer's profit to zero gives the breakeven share price $S^{\ddagger}$ at expiration:

$$S^{\ddagger} = K + \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c(K, S(0)) \qquad \text{call}$$

$$S^{\ddagger} = K - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p(K, S(0)) \qquad \text{put}$$

Above the call breakeven, and below the put breakeven, the buyer profits and the seller loses.

Listed contracts often expire within weeks, so $\mathcal{D}_{T,0}(g_y)$ is close to one (about 1.003 for 30 days at $g_y = 4\%$). For longer-dated contracts it matters: a 1-year contract at $g_y = 4\%$ has $\mathcal{D}_{T,0}(g_y) = e^{0.04} \approx 1.041$, which moves the breakeven farther from the strike.

Example 1: CHEME-5660-L9a-Example-SingleContractPayoffProfit-Fall-2026.ipynb

Pricing Put and Call Options Contracts

An option's price depends on the underlying asset's value, strike price, time to expiration, and exercise style.

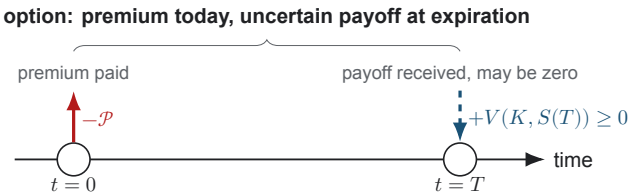
- European-style: exercise only at expiration.
- American-style: exercise any time through expiration.

The American option's added flexibility makes it worth at least as much as an otherwise identical European option.

Options on single stocks and on exchange-traded funds such as SPY are American-style. The European-style contracts we can trade are cash-settled index options: SPX (SPXW for weekly expirations), XSP, NDX, and RUT.

Let's start with European-style options and compute their net present value.

Options as Abstract Assets



The buyer pays the premium $\mathcal{P}$ today and receives the payoff $V(K, S(T)) \geq 0$ at expiration, so the net present value at $t = 0$ is:

$$\text{NPV}_0 = -\mathcal{P} + \mathcal{D}_{T,0}^{-1}(g_y) \cdot V(K, S(T))$$

Factoring out $\mathcal{D}_{T,0}^{-1}(g_y)$ gives $\text{NPV}_0 = \mathcal{D}_{T,0}^{-1}(g_y) \cdot P^{\text{buyer}}$: the NPV is the buyer's profit at expiration discounted to today, so it is zero at the breakeven share price.

The European Call Contract Premium

How is the market price set? A call pays $S(T) - K$ when $S(T) > K$, and nothing otherwise. Following the same steps as for the bet:

$$\begin{aligned}
\mathbb{E}[\mathrm{NPV}_0] &= \left(-\mathcal{P} + \mathcal{D}_{T,0}^{-1}(g_y)\,\bar{V}\right)\mathbb{P}(S(T) > K) + (-\mathcal{P})\,\mathbb{P}(S(T) \le K) && \text{weighted sum} \\
&= -\mathcal{P}\ \underbrace{\left[\mathbb{P}(S(T) > K) + \mathbb{P}(S(T) \le K)\right]}_{=1} + \mathcal{D}_{T,0}^{-1}(g_y)\,\bar{V}\,\mathbb{P}(S(T) > K) && \text{collect } \mathcal{P} \text{ terms, sum to one} \\
&= -\mathcal{P} + \mathcal{D}_{T,0}^{-1}(g_y)\,\mathbb{P}(S(T) > K)\,\bar{V}
\end{aligned}$$

where $\bar{V}$ is the average payoff $S(T) - K$ over the outcomes with $S(T) > K$.

The expected NPV is non-negative when $\mathcal{P} \le \mathcal{D}_{T,0}^{-1}(g_y)\,\mathbb{P}(S(T) > K)\,\bar{V}$: the discounted chance of exercise times the average payoff when exercised.

Whose Probabilities?

The bound on the premium from the last slide depends on whose probabilities we use:

- Our beliefs $\mathbb{P}$: the probabilities are our own beliefs about $S(T)$, so this premium is the most we would pay. A buyer with different beliefs computes a different premium.
- The market's rule $\mathbb{Q}$: the market prices every option with one shared set of probabilities, the risk-neutral pricing rule $\mathbb{Q}$, so there is one price.
- Growth under $\mathbb{Q}$: the expected share price grows at the risk-free rate g_y. For geometric Brownian motion with no dividends, $\mathbb{Q}$ replaces the drift with g_y.

Pricing with $\mathbb{Q}$ gives one market premium that does not depend on who buys the contract.

Expected NPV Under $\mathbb{Q}$

Repeating the expected NPV calculation with $\mathbb{Q}$ in place of $\mathbb{P}$ gives:

$$\begin{aligned}
\mathbb{E}_{\mathbb{Q}}[\mathrm{NPV}_0] &= \left(-\mathcal{P} + \mathcal{D}_{T,0}^{-1}(g_y)\,\bar{V}\right)\mathbb{Q}(S(T) > K) + (-\mathcal{P})\,\mathbb{Q}(S(T) \le K) && \text{weighted sum} \\
&= -\mathcal{P}\,\underbrace{\left[\mathbb{Q}(S(T) > K) + \mathbb{Q}(S(T) \le K)\right]}_{=1} && \text{collect terms} \\
&\quad + \mathcal{D}_{T,0}^{-1}(g_y)\,\underbrace{\left[\bar{V}\,\mathbb{Q}(S(T) > K) + 0 \cdot \mathbb{Q}(S(T) \le K)\right]}_{=\mathbb{E}_{\mathbb{Q}}[V_c(K,S(T))]} && \text{call pays 0 if } S(T) \le K \\
&= -\mathcal{P} + \mathcal{D}_{T,0}^{-1}(g_y)\,\mathbb{E}_{\mathbb{Q}}[V_c(K, S(T))]
\end{aligned}$$

where $\bar{V}$ is now the average payoff $S(T) - K$ over the outcomes with $S(T) > K$ under $\mathbb{Q}$.

The second bracket is the expected payoff of the call under $\mathbb{Q}$: the payoff in each outcome, weighted by its probability.

The Market Premium

The seller's NPV is the negative of the buyer's. The only premium that gives neither side a positive expected NPV under $\mathbb{Q}$ sets $\mathbb{E}_{\mathbb{Q}}[\mathrm{NPV}_0] = 0$:

$$\boxed{\mathcal{P}_c^{\mathsf{EU}} = \mathcal{D}_{T,0}^{-1}(g_y)\,\mathbb{E}_{\mathbb{Q}}[V_c(K, S(T))]}$$

A put, which pays $K - S(T)$ when $S(T) < K$, gives the same result with V_p in place of V_c. For either contract, with payoff $V_\star(K, S(T))$:

$$\boxed{\mathcal{P}_\star^{\mathsf{EU}} = \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{E}_{\mathbb{Q}}[V_\star(K, S(T))] \quad \text{where } \star \in \{c, p\}}$$

The European premium is the discounted expected payoff under $\mathbb{Q}$, one price for all buyers. Monte Carlo simulation and the Black–Scholes–Merton formulas are two ways to compute it.

A Monte Carlo Premium

Simulation estimates the market premium directly. The payoff depends only on $S(T)$, so we draw it from the solution of geometric Brownian motion (GBM) with drift g_y and volatility σ.

Initialize: given $S(0)$, K, the payoff $V(K, S)$, T (years), σ, g_y, and M paths.

1. For each path $i = 1, 2, \ldots, M$ **do**:
   a. Draw a standard normal random number Z_i.
   b. $S_i(T) \leftarrow S(0) \cdot \exp\left[(g_y - \sigma^2/2)\,T + \sigma\sqrt{T}\,Z_i\right]$.
   c. $V_i \leftarrow V(K, S_i(T))$.
2. Discount the average payoff: $\mathcal{P} \leftarrow \mathcal{D}_{T,0}^{-1}(g_y) \cdot \frac{1}{M} \sum_{i=1}^{M} V_i$.

The result is an estimate: it changes with the random draws, and it becomes more precise as M grows.

Optional: an alternative way to think about the premium, in the contingent claims notebook.

The Black–Scholes–Merton Equation

Merton and Scholes received the 1997 Nobel Prize in Economic Sciences for this model. Fischer Black had passed away before the award.

Assume the price of a non-dividend-paying share follows GBM with constant σ and a constant, continuously compounded risk-free rate g_y. The premium $\mathcal{P}(S,t)$, with $\mathcal{P}(S,T) = V(K,S)$, satisfies:

$$\boxed{\frac{\partial \mathcal{P}}{\partial t} + g_y S \frac{\partial \mathcal{P}}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 \mathcal{P}}{\partial S^2} = g_y \mathcal{P}}$$

Solving it backward in time from the payoff gives today's premium, $\mathcal{P}(S(0),0)$, and this solution equals the risk-neutral premium $\mathcal{P}^{\text{EU}}$ that the Monte Carlo simulation estimates.

Derivation: the contingent claims notebook.

Black–Scholes–Merton Premiums

Instead of simulating, we can compute this premium with the closed-form European call and put formulas:

$$\boxed{\begin{aligned}\mathcal{P}_c(K, S(0)) &= N(d_+) \cdot S(0) - N(d_-) \cdot K \cdot \mathcal{D}_{T,0}^{-1}(g_y) \\ \mathcal{P}_p(K, S(0)) &= N(-d_-) \cdot K \cdot \mathcal{D}_{T,0}^{-1}(g_y) - N(-d_+) \cdot S(0)\end{aligned}}$$

where N is the standard normal CDF, T is in years, and:

$$d_+ = \frac{1}{\sigma\sqrt{T}}\left[\ln\left(\frac{S(0)}{K}\right) + \left(g_y + \frac{\sigma^2}{2}\right)T\right], \qquad d_- = d_+ - \sigma\sqrt{T}$$

We know $S(0)$, K, T, and g_y when we price the contract. The volatility σ is the one input we must choose.

Which Volatility Should We Use?

If we knew the right volatility, the BSM model (or the Monte Carlo model) would compute the observed market premium. In practice, we observe the market premium and back out the volatility that makes the model match it.

- Historical volatility: $\sigma_{\text{ann}} = \sqrt{\Delta t} \cdot \sigma_g$, where σ_g is the standard deviation of past growth rates (units: yr^{-1}) and Δt is the time between observations (units: years). It looks backward.
- Implied volatility (IV): the market's forecast of a likely future movement in the share price. It looks forward, and it is often at or above the historical volatility.
- How can we use it? IV represents a one-standard-deviation movement in the share price over the next year. Over T years, the move scales with $\sqrt{T}$.

This $\sqrt{T}$ scaling gives a quick range for the share price on the next slide.

A Back-of-the-Envelope Range from IV

Moving one standard deviation up or down in the solution of geometric Brownian motion, $Z = \pm 1$, gives a back-of-the-envelope range for the share price:

$$
\begin{aligned}
S(T) &= S(0)\exp\left[\mu_g T + \sigma\sqrt{T}\,Z\right] && \text{solution of geometric Brownian motion} \\
S_{\pm} &= S(0)\exp\left[\mu_g T \pm \mathsf{IV}\sqrt{T}\right] && \text{set } \sigma = \mathsf{IV} \text{ and } Z = \pm 1 \\
&\approx S(0)\exp\left[\pm \mathsf{IV}\sqrt{T}\right] && \mu_g T \text{ is small next to } \mathsf{IV}\sqrt{T} \text{ for short } T \\
&\approx S(0)\left(1 \pm \mathsf{IV}\sqrt{T}\right) && e^x \approx 1 + x \text{ for small } x
\end{aligned}
$$

where μ_g is the mean growth rate, Z is standard normal, and $S_{\pm}$ is the share price one standard deviation above or below $S(0)$. Under the model, the share price ends inside this range about 68% of the time.

Implied Volatility by Bisection

We back out the implied volatility $\hat{\sigma}$ by bisection.

Initialize: given $\mathcal{P}^{\text{market}}$, $S(0)$, K, T (years), g_y, a bracket $\sigma_{\text{lo}} < \sigma_{\text{hi}}$ with $\mathcal{P}(\sigma_{\text{lo}}) \le \mathcal{P}^{\text{market}} \le \mathcal{P}(\sigma_{\text{hi}})$, a tolerance $\epsilon > 0$, `maxiter`, and $i \leftarrow 1$.

While $\sigma_{\text{hi}} - \sigma_{\text{lo}} > \epsilon$ and $i \le$ `maxiter` **do**:

1. $\sigma_{\text{mid}} \leftarrow (\sigma_{\text{lo}} + \sigma_{\text{hi}})/2$ and compute its BSM premium $\mathcal{P}(\sigma_{\text{mid}})$.
2. If $\mathcal{P}(\sigma_{\text{mid}}) < \mathcal{P}^{\text{market}}$, **then** $\sigma_{\text{lo}} \leftarrow \sigma_{\text{mid}}$. Otherwise, $\sigma_{\text{hi}} \leftarrow \sigma_{\text{mid}}$.
3. $i \leftarrow i + 1$.

The implied volatility is the midpoint of the final bracket, $\hat{\sigma} \leftarrow (\sigma_{\text{lo}} + \sigma_{\text{hi}})/2$. The BSM premium rises with σ, so exactly one σ matches the market premium, and each step halves the bracket around it.

Implied Volatility in Practice

Three points about using implied volatility with real quotes:

- Market quotes: the SPXW volatility skew notebook applies this bisection to a chain of S&P 500 index options.
- One per contract: each strike and expiration has its own implied volatility, although the BSM model assumes one constant σ. This pattern is known as the volatility smile or volatility skew.
- Pricing without a quote: to price a contract without a quote, or on a future date, we can predict its implied volatility from a surface fitted to today's quotes, which gives implied volatility as a function of strike and expiration (Sun et al., 2026).

Next, we link the call and put premiums with the same strike and expiration.

Optional: the SPXW volatility skew notebook.

Put–Call Parity: The Payoff Identity

The call and put premiums are linked. For a European call and put with the same strike and expiration, the payoffs differ by $S(T) - K$ at every share price:

$$
\begin{aligned}
V_c(K, S(T)) - V_p(K, S(T)) &= \max(S(T) - K,\ 0) - \max(K - S(T),\ 0) && \text{payoff definitions} \\
&= \begin{cases} (S(T) - K) - 0 & \text{if } S(T) > K\text{: put pays 0} \\ 0 - (K - S(T)) & \text{if } S(T) \le K\text{: call pays 0} \end{cases} && \text{split at } K \\
&= S(T) - K && \text{simplify both cases}
\end{aligned}
$$

Because the payoffs differ by $S(T) - K$ in every outcome, the premiums differ by the discounted expected value of $S(T) - K$ under $\mathbb{Q}$.

Put–Call Parity for European Contracts

Subtracting the put premium from the call premium gives:

$$
\begin{aligned}
&\mathcal{P}_c(K, S(0)) - \mathcal{P}_p(K, S(0)) \\
&\quad = \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{E}_{\mathbb{Q}}[V_c(K, S(T))] - \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{E}_{\mathbb{Q}}[V_p(K, S(T))] && \text{premium formula} \\
&\quad = \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{E}_{\mathbb{Q}}[V_c(K, S(T)) - V_p(K, S(T))] && \text{expectation is linear} \\
&\quad = \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{E}_{\mathbb{Q}}[S(T) - K] && \text{payoff identity} \\
&\quad = \mathcal{D}_{T,0}^{-1}(g_y) \cdot (S(0)\, e^{g_y T} - K) && \text{expected share price grows at } g_y \\
&\quad = S(0) - K \cdot \mathcal{D}_{T,0}^{-1}(g_y) && \mathcal{D}_{T,0}^{-1}(g_y) = e^{-g_y T}
\end{aligned}
$$

This is put–call parity: $\boxed{\mathcal{P}_c(K, S(0)) - \mathcal{P}_p(K, S(0)) = S(0) - K \cdot \mathcal{D}_{T,0}^{-1}(g_y)}$

The call–put difference does not depend on σ. It depends only on $S(0)$, K, and the discount factor.

Example 2: CHEME-5660-L9a-Example-BSM-Premium-Fall-2026.ipynb

What About American-Style Options?

Put–call parity holds only for European contracts. The derivation compares the two payoffs on the expiration date, but an American holder can exercise before then. For American contracts, the difference between the call and put premiums is bounded, not fixed.

The American holder can exercise at any time $\tau \leq T$, chosen using only the information available at that time. The premium is the largest expected payoff, discounted from τ back to today, over all such exercise times:

$$\mathcal{P}_\star^{\mathsf{AM}} = \max_{\tau \leq T} \mathbb{E}_{\mathbb{Q}}\left[\mathcal{D}_{\tau,0}^{-1}(g_y) \cdot V_\star(K, S(\tau))\right] \geq \mathcal{P}_\star^{\mathsf{EU}} \quad \text{where } \star \in \{c, p\}$$

The American put has no closed-form formula like BSM, so its premium is usually computed on a binomial lattice.

Optional Advanced Material

Two optional notebooks extend this lecture:

Contingent claims: advanced/contingent_claims/
CHEME-5660-L9a-Advanced-ContingentClaims-BSM-Fall-2026.ipynb

We derive where the BSM premiums come from: the risk-neutral probability on a one-step lattice, state prices for event contracts, the BSM equation and its solution, and the call built from a digital claim and an asset-or-nothing claim. We also compare a client's certainty equivalent for the bet with its market price.

Volatility skew: advanced/spxw_volatility_skew/
CHEME-5660-L9a-Advanced-SPXW-Volatility-Skew-Fall-2026.ipynb

We apply the implied-volatility bisection to an end-of-day chain of SPXW options and compare the skew with the realized volatility of SPY, which tracks the index.

Notebooks: lectures/week-9/L9a/advanced, also linked from the lecture notebook.

Summary

In this lecture, we explored derivatives, a fundamental asset class used for risk management, hedging, and speculation.

Key takeaways

- Derivatives derive their value from underlying assets: common types are futures, forwards, swaps, and options, each with different characteristics, maturities, and uses.
- Option payoffs are determined by strike prices and contract orientation: payoff diagrams at expiration show profitability and risk exposure.
- The premium makes the expected NPV zero under the risk-neutral pricing rule: Monte Carlo simulation or the BSM formulas compute it for European contracts, and put–call parity links calls and puts.

Understanding derivative mechanics, payoff structures, and pricing is important for financial decision-making.